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Sleep Disorder Diagnosis Using Machine Learning

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Abstract:	Not getting proper sleep is a serious issue that affects both physical health and mental well-being. Poor sleep can lead to problems such as reduced concentration, mood disorders, weakened immunity, and long-term health complications. However, many people remain unaware that they are suffering from a sleep disorder because traditional diagnostic methods, such as sleep studies, are often expensive, time-consuming, and not easily accessible. Moreover, these methods may not always capture early-stage symptoms or subtle patterns in daily behaviour. This project focuses on developing an AI-based system capable of detecting sleep disorders using simple and easily available inputs such as age, heart rate, daily habits, and lifestyle factors. The aim is to create a solution that is both practical and scalable, allowing early identification of potential sleep-related issues without the need for complex medical procedures. To achieve this, the dataset was carefully pre-processed by handling missing values, removing inconsistencies, and normalizing features to improve model performance. Multiple machine learning algorithms were implemented and compared to identify the most effective model for accurate prediction. Evaluation metrics such as accuracy, precision, recall, and F1-score were analysed to ensure the reliability and robustness of the system. In addition, explainable AI techniques were incorporated to provide transparency in the model's decision-making process. This helps in understanding why a particular prediction is made, reducing the "black box" nature of AI systems and increasing trust among healthcare professionals. The goal of this project is to develop a user-friendly and efficient tool that can assist in the early detection of sleep disorders, enabling timely intervention and better health outcomes. By simplifying the diagnostic process and making it more accessible, this system has the potential to significantly improve the quality of life for individuals affected by sleep-related issues.

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Date: 02/04/2026

To
The Editor
Journal of Medical Imaging and Radiation Sciences

Subject: Submission of Manuscript for Consideration

Dear Editor,

I am pleased to submit our manuscript titled “Sleep Disorder Diagnosis Using Machine Learning” for consideration for publication in your esteemed Scopus-indexed journal.

This manuscript presents a novel machine learning–based framework for the early detection and classification of sleep disorders using easily accessible demographic, behavioral, and physiological data. The study addresses the limitations of traditional diagnostic approaches such as polysomnography, which are often expensive, time-consuming, and inaccessible to a large population.

In this work, we developed a structured multi-stage system that includes data preprocessing, feature extraction, model training, and classification. Multiple machine learning algorithms were evaluated using performance metrics such as accuracy, precision, recall, and F1-score to ensure robustness and reliability. Additionally, explainable AI techniques have been incorporated to enhance transparency and interpretability of the model’s predictions, making the system more trustworthy for healthcare applications.

The proposed approach contributes to the growing field of AI-driven healthcare by providing a scalable, cost-effective, and user-friendly solution for early diagnosis of sleep disorders. We believe that this work aligns well with the scope of your journal and will be of significant interest to researchers and practitioners in machine learning, healthcare analytics, and biomedical engineering.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to your journal.

We have no conflicts of interest to disclose. All ethical standards have been followed during the preparation of this work.

Thank you for considering our submission. We look forward to your positive response.

Sincerely,

Vishal Kumar Yadav

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Sleep Disorder Diagnosis Using Machine Learning

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Keywords: Sleep disorders, prediction model, machine learning, diagnosis, features, classification, healthcare.

1. INTRODUCTION

Sleep has long been recognized as a vital component of human health, directly influencing cognitive function, metabolic regulation, emotional stability, and overall quality of life. However, in recent years, the prevalence of sleep disorders has risen sharply due to modern lifestyle disruptions, increased stress levels,

irregular work routines, and prolonged exposure to digital devices. Conditions such as insomnia, obstructive sleep apnea, hypersomnia, and circadian rhythm disruptions significantly impair productivity, weaken immune response, and increase the risk of chronic illnesses including cardiovascular diseases, hypertension, obesity, and diabetes. Despite the seriousness of these conditions, sleep disorders often go undetected or misdiagnosed because early symptoms are subtle, and traditional diagnostic methods remain complex, time-intensive, and costly.

Conventional diagnosis primarily relies on **polysomnography (PSG)**, considered the gold standard for sleep evaluation. PSG involves overnight monitoring of physiological signals such as brain activity, respiratory cycles, muscle movements, heart rate, and blood oxygen levels. Although accurate, PSG presents substantial challenges: it requires specialized laboratories, skilled technicians, high-cost equipment, and lengthy analysis time. Additionally, the controlled clinical environment can alter a patient’s natural sleep behavior, affecting the accuracy of the recorded patterns. These limitations restrict accessibility, especially in rural or resource-constrained regions, leaving a significant portion of affected individuals without timely diagnosis and intervention.

Traditional manual assessments also fail to address the **heterogeneity** and **dynamic nature** of sleep-related behaviors. Sleep characteristics vary widely across age groups, occupations, health conditions, and lifestyle patterns, making it difficult to apply a one-size-fits-all diagnostic approach. With the widespread adoption of wearable devices and health-tracking applications, individuals now generate large volumes of sleep-related data, including heart rate, physical activity levels, step count, sleep duration, and stress level. However, analyzing this high-dimensional data manually is impractical and prone to inconsistencies, highlighting the need for automated, intelligent systems capable of deriving meaningful insights quickly and accurately.

Recent advancements in **machine learning (ML)** have created new opportunities for developing automated diagnostic frameworks that can predict sleep disorders using easily measurable features. ML algorithms excel at learning nonlinear relationships, identifying complex patterns, and detecting subtle deviations in behavioral and physiological indicators. By training predictive models on large, labeled datasets, it becomes possible to classify individuals

into sleep disorder categories with high accuracy, without requiring intrusive clinical procedures. Techniques such as Random Forest, Support Vector Machine (SVM), Logistic Regression, Gradient Boosting, and K-Nearest Neighbors (KNN) have shown promising results in medical classification tasks, making them suitable candidates for sleep disorder prediction.

However, despite these advancements, existing solutions still face limitations. Many research models depend on narrowly focused datasets or limited feature sets, reducing generalization across diverse populations. Data collected from consumer-grade wearables may contain noise, missing values, or inconsistencies that complicate the training process. Privacy and security concerns also emerge when storing sensitive health data, requiring robust anonymization and ethical data-handling strategies. These challenges emphasize the need for a more comprehensive, reliable, and adaptable machine learning framework capable of functioning effectively across real-world environments.

As healthcare trends shift toward **preventive monitoring** and **early detection**, machine learning–based diagnostic tools offer a transformative solution. Such systems reduce dependency on specialized sleep laboratories and enable large-scale, low-cost screening. When integrated into smartphones, IoT-based wearables, and digital health platforms, predictive models can provide real-time alerts, personalized recommendations, and continuous monitoring, empowering individuals to take proactive control of their sleep health.

This research proposes an ML-based predictive framework designed to classify sleep disorders using demographic, behavioural, and physiological features. The approach integrates data preprocessing, feature analysis, supervised learning, and performance evaluation to determine the most effective algorithm for multi-class prediction. By offering a scalable and accessible diagnostic method, the proposed system contributes to overcoming existing limitations and supports improved healthcare accessibility, early detection, and better management of sleep-related disorders.

2. LITERATURE REVIEW

Sleep disorder research has gained significant attention over the past two decades due to the increasing global prevalence of sleep-related illnesses and their strong correlation with chronic health conditions. Early studies primarily focused on clinical methodologies, especially polysomnography (PSG), which remains the gold standard for diagnosing sleep disorders. However, with the rise of artificial intelligence (AI), machine learning (ML), and wearable technologies, researchers have shifted toward automated, scalable, and cost-effective diagnostic approaches. This literature review discusses major findings, methodologies, and limitations across existing studies related to sleep pattern analysis, ML-based classification, sensor data processing, and health monitoring systems.

Early research in sleep disorder diagnosis concentrated on understanding the physiological basis of conditions such as insomnia, obstructive sleep apnea (OSA), and hypersomnia. Rechtschaffen and Kales (R&K) established one of the earliest frameworks for sleep stage scoring based on EEG, EOG, and EMG signals. Their rule-based manual scoring system laid the foundation for PSG-based diagnosis but required substantial human expertise and time. Though accurate, such manual evaluation methods lacked scalability, motivating further research into automated classification systems.

Advancements in biomedical signal processing led to attempts at automating PSG data interpretation. Researchers initially applied signal decomposition, spectral analysis, and statistical modeling to EEG signals for detecting sleep stages. For example, Rechtschaffen's successors proposed the use of Fourier analysis to differentiate between REM and NREM stages. While these approaches improved accuracy, the complexity and variability of physiological signals limited the effectiveness of traditional statistical models. Consequently, machine learning techniques began emerging as promising alternatives.

Supervised ML models such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), Decision Trees, and Random Forests gained traction due to their ability to classify high-dimensional biomedical data. Shambroom et al. explored automated sleep staging using SVM to categorize PSG signals into different sleep phases. Their findings indicated that SVM performed well in detecting patterns but required extensive feature engineering to achieve optimal results. Similarly, Naïve Bayes and KNN were applied to early sleep datasets, showing reasonable accuracy but struggling with data imbalance and nonlinear relationships between features.

As researchers recognized the importance of behavioral and lifestyle factors in sleep disorders, attention shifted toward datasets incorporating variables such as BMI, physical activity, sleep duration, stress levels, and heart rate. Buysse et al. introduced the Pittsburgh Sleep Quality Index (PSQI), a self-reported measure that correlates lifestyle habits with sleep quality. While subjective, PSQI significantly influenced later studies that began integrating health metrics with machine learning to predict sleep conditions.

Recent work has focused heavily on using wearable and mobile technologies to collect physiological signals. Fitbit, Apple Watch, and other consumer-grade devices generate continuous data streams, enabling large-scale, real-time sleep monitoring. De Zambotti et al. evaluated the reliability of wearable sensors and concluded that although they do not match PSG-level precision, they provide valuable approximations suitable for ML-based predictive models. Such findings encouraged researchers to combine wearable sensor data with ML techniques to detect sleep disorders.

Deep learning approaches have gained popularity in biomedical research due to their ability to learn complex data representations. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have been applied to raw EEG and respiratory data. For instance, Supratak et al. introduced DeepSleepNet, a CNN-LSTM hybrid model for sleep stage classification using raw single-channel EEG data. Their model significantly outperformed traditional ML approaches, demonstrating the potential of deep learning in automated sleep assessment. However, deep learning models required large datasets, high computational resources, and longer training times, limiting their accessibility.

In the context of sleep apnea detection, researchers have applied ML and deep learning on respiratory signals, oxygen saturation levels, and heart rate variability (HRV). Almazaydeh et al. utilized logistic regression and decision trees to classify apnea from ECG-derived features. Their work demonstrated the usefulness of ML but also highlighted challenges with low signal quality and data generalization across patients. Further research by Liang et al. applied CNNs to SpO_2 signals, showing improved detection accuracy albeit with higher computational demands.

Another branch of research investigated the psychological and environmental aspects influencing sleep. Studies by Sadeh and colleagues explored the role of stress, daily routines, and emotional states in determining sleep quality. These studies formed the basis for multi-factor ML models

that combine physiological and psychological features. For instance, a study by Liu et al. integrated wearable sensor data with self-reported stress scores to predict insomnia. Their findings suggested that hybrid models incorporating both objective and subjective features yield higher predictive accuracy.

The integration of ML with Internet of Things (IoT) technologies further expanded the scope of sleep disorder research. IoT-based approaches allow continuous monitoring through sensors embedded in mattresses, smart home devices, and mobile applications. Hassan et al. proposed an IoT-based sleep monitoring framework that collects sleep posture, temperature, and movement data for classification using Random Forest and Gradient Boosting. Their results highlighted the potential for real-time monitoring but also exposed concerns related to data privacy, security, and device reliability.

Several comparative studies examined the performance of different ML algorithms for sleep classification. Random Forest consistently emerged as a strong performer due to its ability to handle nonlinear relationships, noisy inputs, and imbalanced classes. SVM, while effective in high-dimensional spaces, required extensive tuning and performed poorly with large datasets. Logistic Regression provided interpretability but lacked the complexity needed for multi-class classification. Gradient Boosting methods such as XGBoost showed high accuracy but involved long training periods. These findings underscore the importance of choosing appropriate models based on dataset size, complexity, and real-time requirements.

A recent trend involves multimodal learning, where sleep prediction models integrate multiple data sources such as motion sensors, heart rate monitors, microphones, and environmental sensors. Multi-sensor fusion improves diagnostic accuracy but increases the complexity of data preprocessing and model training. Studies by Fernandez et al. demonstrated that combining HRV, actigraphy, and respiratory signals enhances sleep apnea detection. However, limitations related to synchronization, missing data, and computational overhead remain significant barriers.

Ethical, privacy, and reliability challenges have also been widely discussed in the literature. Researchers noted that health data collected from wearables can be inaccurate or biased due to device limitations. Concerns related to data security and compliance with healthcare privacy regulations such as HIPAA and GDPR are also relevant. Studies emphasize the need for anonymization, secure data handling, and transparent model interpretation when implementing ML-based sleep diagnostic systems.

Overall, the literature reveals several important insights. First, ML-based sleep disorder prediction demonstrates strong potential, especially when integrating physiological, behavioural, and environmental data. Second, no single model universally outperforms others; instead, model effectiveness depends on the feature set, dataset characteristics, and intended clinical application. Third, recent advancements in deep learning and IoT technologies have considerably improved predictive accuracy, but present challenges related to resource requirements and privacy.

3.EXISTING SYSTEM

The Existing systems for diagnosing sleep disorders primarily rely on **manual clinical evaluation** and **overnight polysomnography (PSG)** conducted in sleep laboratories. PSG measures multiple physiological parameters—such as brain activity, oxygen saturation, muscle

tone, breathing rate, and heart rhythm—but it is time-consuming, expensive, and requires specialized equipment and trained medical staff. Many hospitals still depend on **subjective patient questionnaires**, sleep diaries, and physical examinations, which often lead to inconsistent or delayed diagnoses. Traditional statistical methods used for analysis are limited in handling large-scale, multi-dimensional data. As a result, early detection becomes challenging, and many patients remain undiagnosed due to high costs, accessibility issues, and long waiting periods.

4.PROPOSED SYSTEM

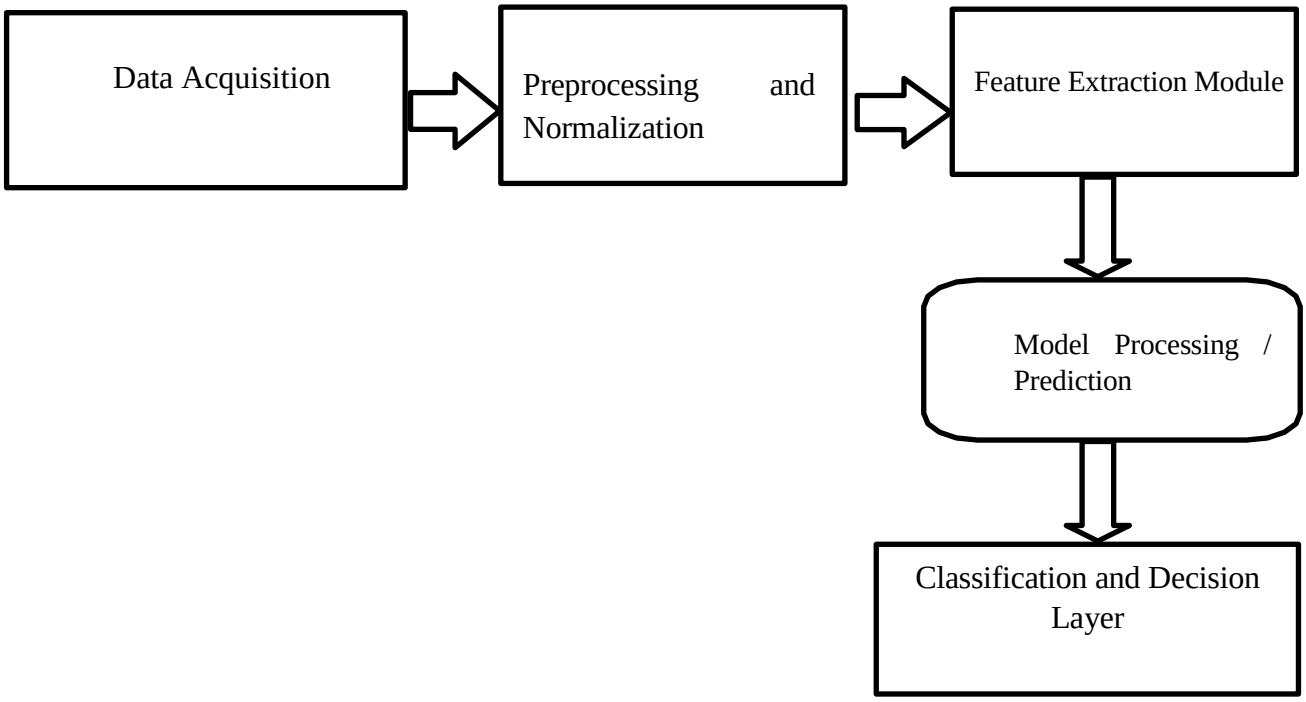

Fig 4.1: System Flow Diagram

The proposed system introduces a structured multi-stage workflow designed to enhance accuracy, reduce processing overhead, and ensure consistent performance. The architecture follows a sequential flow where each module independently contributes to the final decision-making process. The system begins with data acquisition, followed by preprocessing to remove noise and standardize inputs. Relevant features are then extracted to reduce complexity and focus on significant attributes. The processed features are passed to the model layer, where analytical algorithms generate predictions. These outcomes are evaluated and classified to produce results. Performance metrics are continuously monitored to refine subsequent iterations.

A. Data Acquisition

This module collects raw input data required for system processing. It gathers relevant parameters from designated sources, ensuring accurate and complete data capture. The acquired data forms the initial input for subsequent preprocessing, feature extraction, and model analysis, enabling reliable and consistent system performance.

B. Preprocessing and Normalization

This module is responsible for cleaning and preparing raw data to ensure it is suitable for further processing. It removes noise and irrelevant information that could negatively impact model accuracy. Missing values are handled using appropriate techniques such as imputation or removal to maintain data integrity. The module also standardizes data formats, ensuring consistency across all input features. Additionally, normalization is applied to scale numeric attributes into a uniform range, improving comparability between features. These preprocessing steps significantly enhance model performance, reduce variability, and ensure more reliable feature extraction and classification results.

C. Feature Extraction Module

The Feature Extraction Module identifies significant attributes from the preprocessed dataset to enhance model performance and reduce computational complexity. It analyzes correlations, removes redundant variables, and transforms raw inputs into meaningful, structured feature vectors. By isolating essential patterns and behavioral indicators, this module ensures that only the most informative data contributes to model training and prediction. Effective feature extraction improves classification accuracy, accelerates processing, and supports robust decision-making within the proposed system architecture.

D. Model Processing / Prediction

This module applies analytical algorithms to processed features, generating predictive outputs that drive the system's functionality. It leverages trained models to identify complex patterns and relationships within the data. Based on these learned patterns, it executes predefined decision rules to ensure consistent and reliable outcomes. The module then produces classification results that help in categorizing inputs accurately. These results play a crucial role in enabling precise and efficient system-level decision-making.

E. Classification and Decision Layer

This module interprets the outputs generated by the predictive model and maps each input to its appropriate category. It carefully analyses the confidence scores associated with each prediction to determine the level of certainty. Based on predefined thresholds and decision rules, it refines and validates the predicted outcomes. The module also handles ambiguous cases by applying additional logic to ensure consistency and accuracy. Finally, it produces the confirmed classification results, supporting reliable and effective system-level decision-making.

Detailed Performance Metrics

Model	Test Accuracy	CV Score (5-fold)	Variance
Random Forest	93.33%	76.49%	16.84%
Gradient Boosting	93.33%	69.56%	23.78%
SVM	92.00%	85.80%	6.20%
KNN	92.00%	70.87%	21.13%
Logistic Regression	90.67%	83.67%	7.00%
Neural Network	93.33%	71.05%	22.29%
AdaBoost	90.67%	88.22%	2.44%
Ensemble (Best)	93.33%	75.69%	17.64%

Fig.4.2: Comparison Table

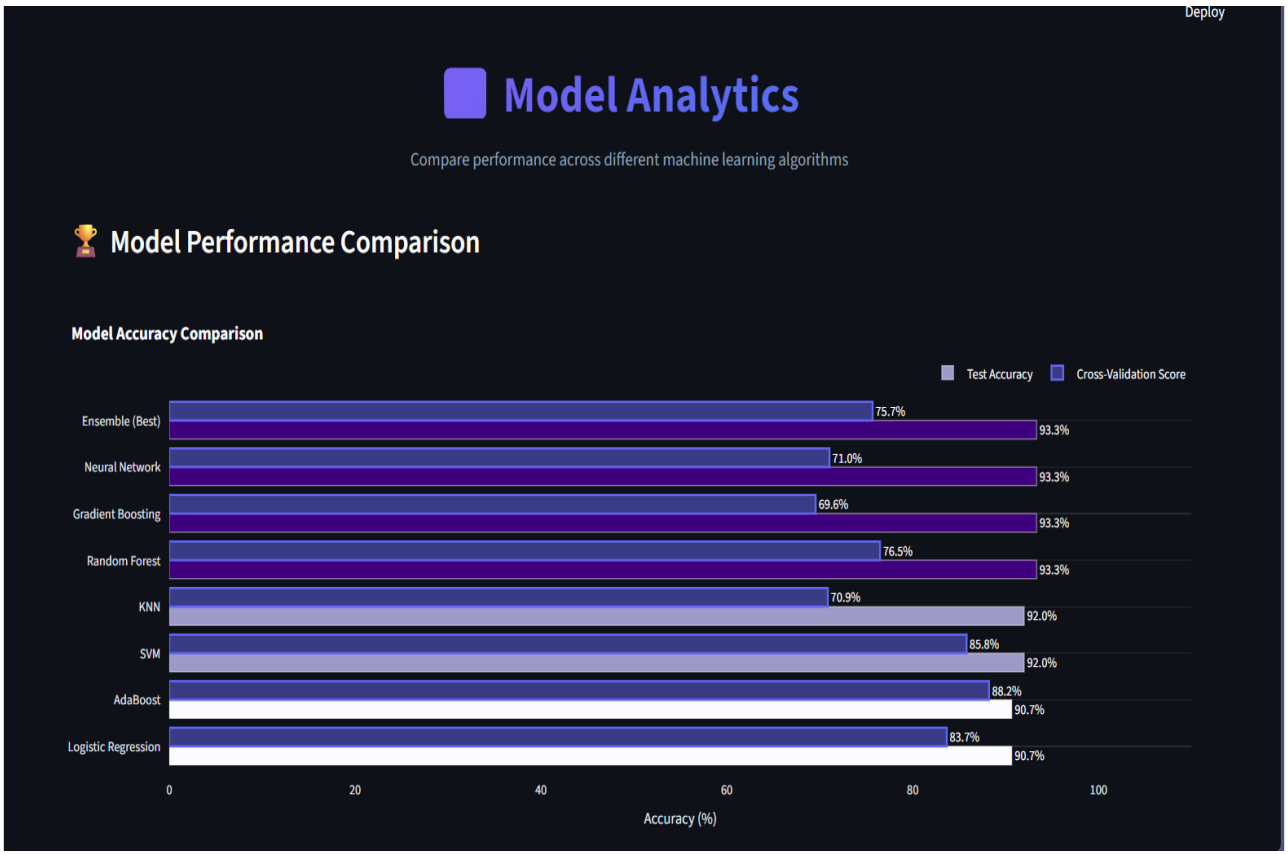

Fig.4.3: Comparison Graph

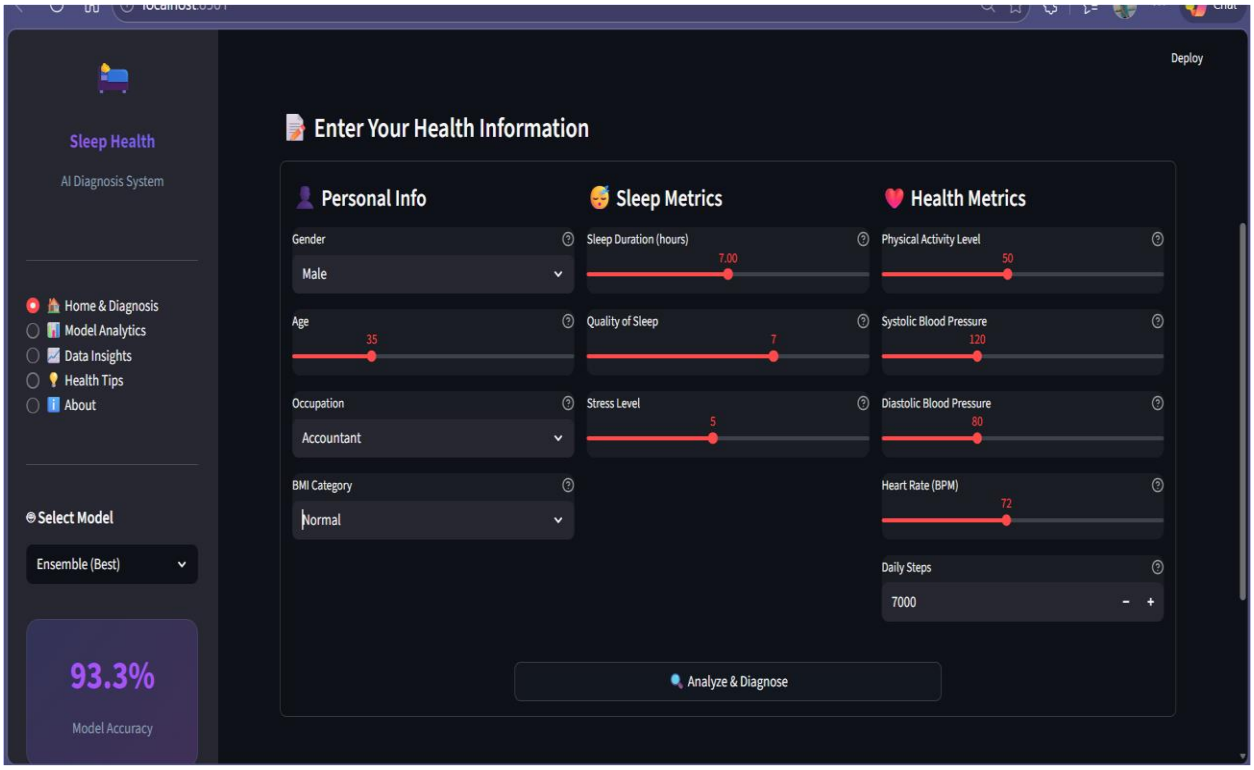

Fig.4.4: Input UI

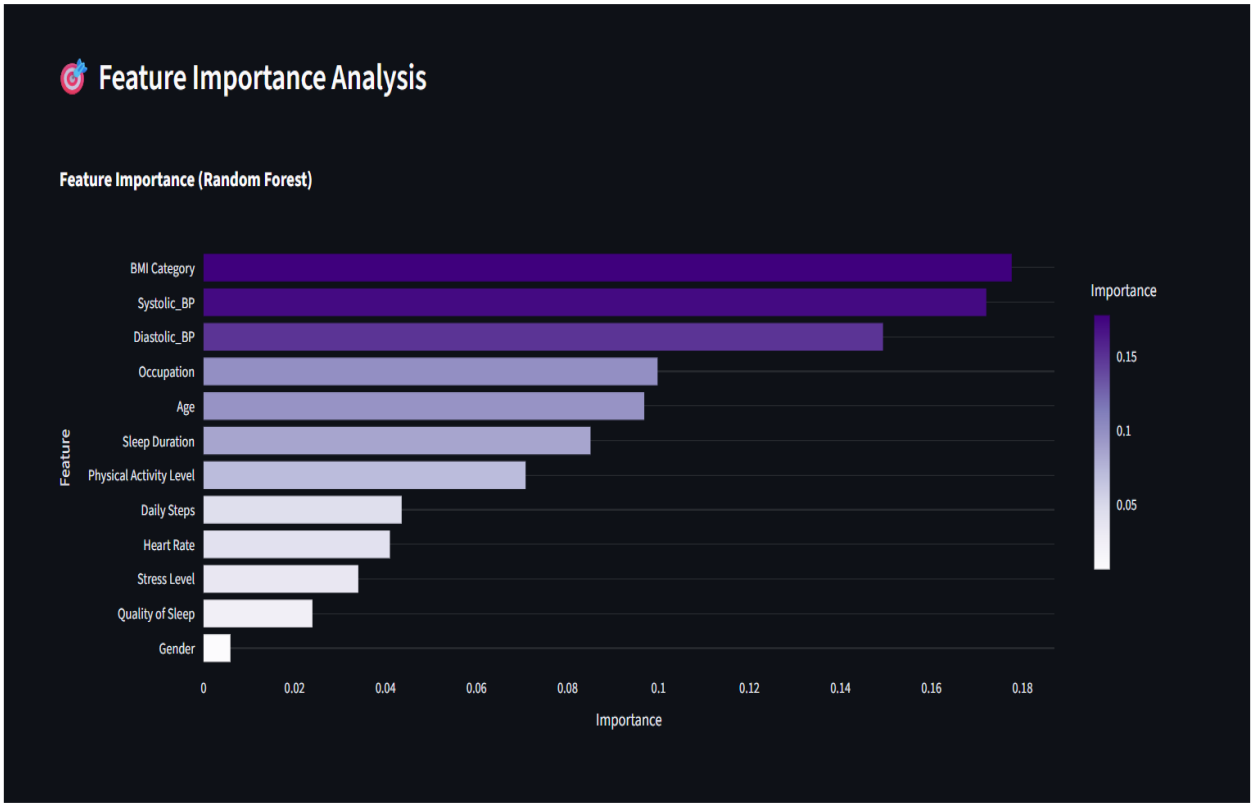

Fig.4.5: Feature Analysis

Fig.4.5: Quality, Occupation

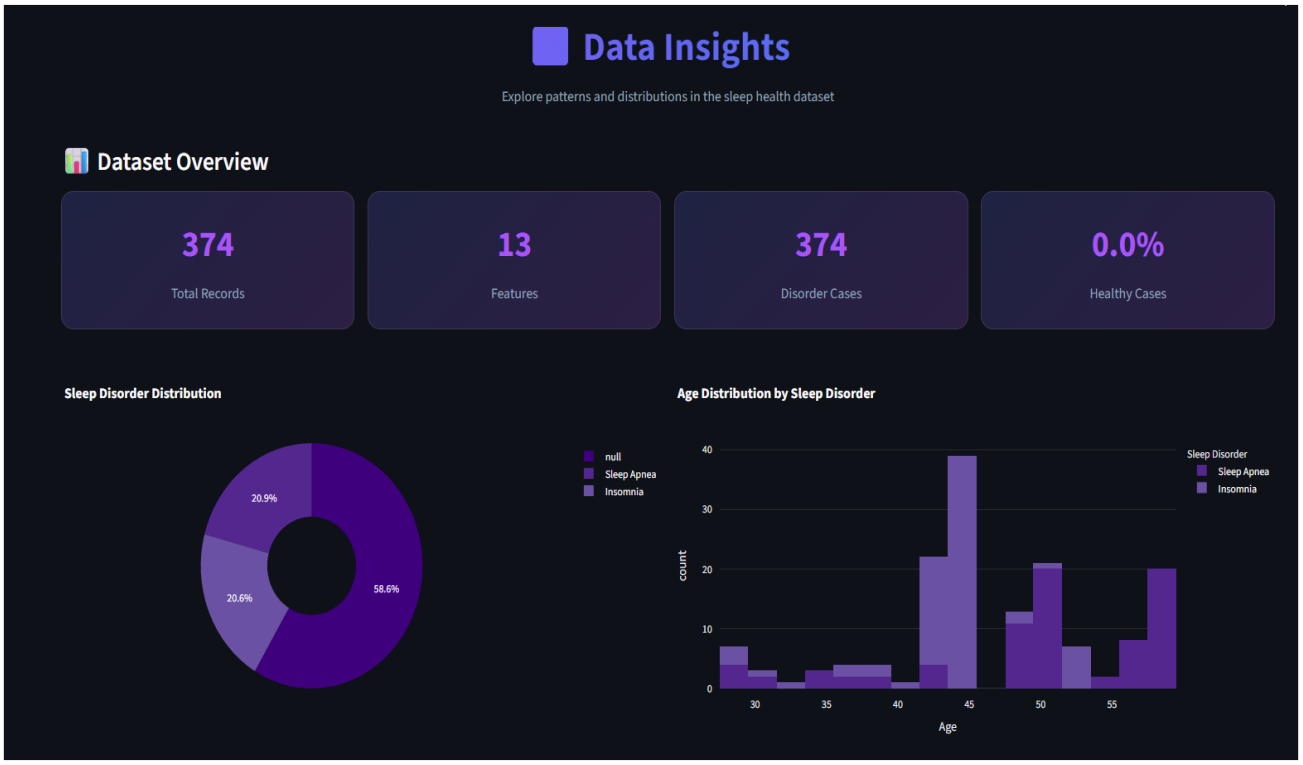

Fig.4.6: Data Insights

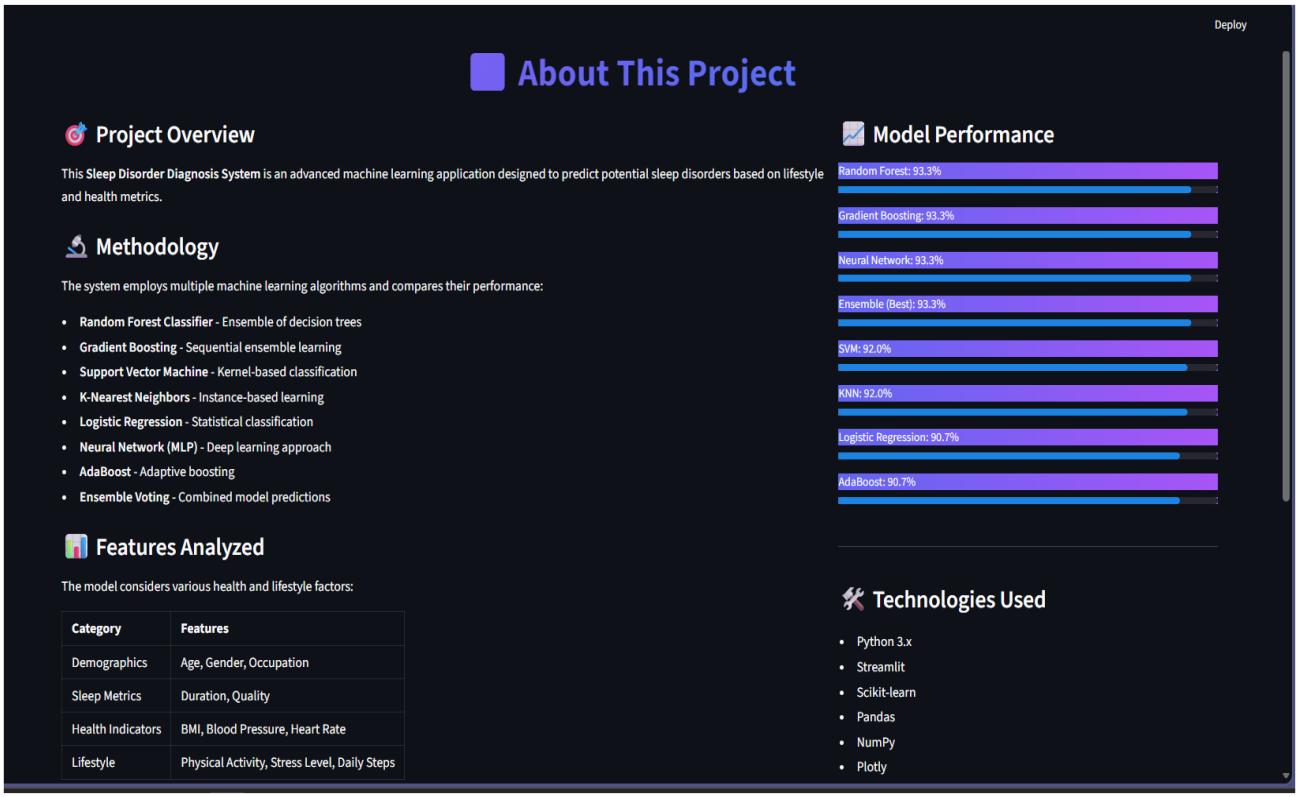

Fig.4.7: About the Project

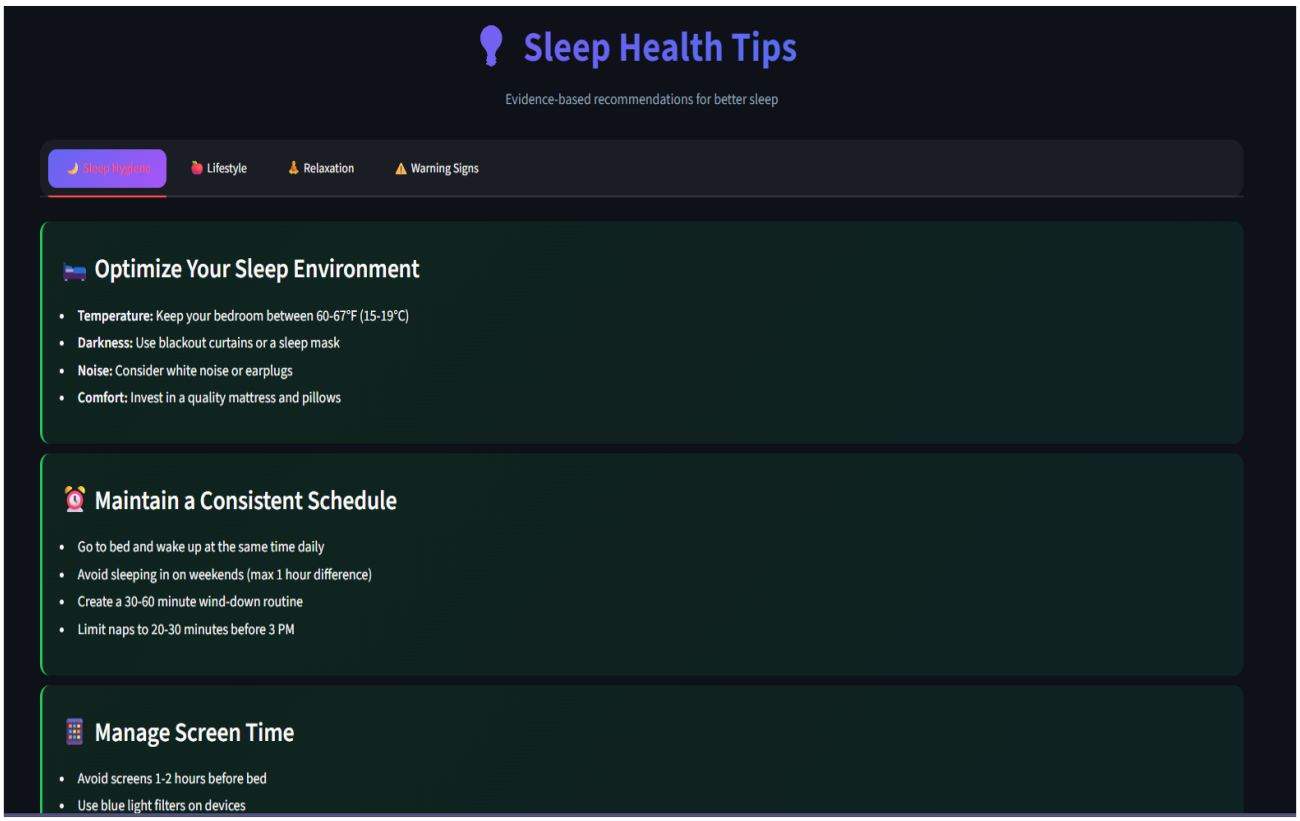

Fig.4.8: Tips

5. CONCLUSION

In This project demonstrates machine learning's strong potential to transform sleep disorder diagnosis. By leveraging algorithms to analyze complex data, from clinical PSG to wearable signals, we can overcome the limitations of traditional methods. Our proposed system facilitates early, accessible detection by creating a scalable screening tool. While challenges like data privacy and model interpretability remain, this approach promises to make sleep disorder assessment more efficient and widespread. Ultimately, it paves the way for timely interventions and improved patient outcomes, marking a significant step forward in sleep medicine.

6. FUTURE WORK

Future work will focus on enhancing the model's robustness and clinical applicability. This includes expanding the dataset with multi-source, multi-center PSG and wearable data to improve generalizability across diverse populations. We will explore advanced deep learning architectures, such as Transformer networks, for superior temporal context modeling. Integrating Explainable AI (XAI) techniques will be prioritized to build clinician trust by providing transparent decision rationales. Furthermore, developing a real-time, edge-computing framework for continuous at-home monitoring and creating a user-friendly clinical decision support system for seamless integration into diagnostic workflows are critical next steps. Finally, longitudinal studies will be essential to validate the model's predictive power for early-onset disorder detection.

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